CHAPTER QUESTIONS

SKILLS REASONING 7

1 State what is formed at the cathode and at the anode during the electrolysis of the following substances. Assume that carbon electrodes were used each time.

a Molten lead(II) bromide

b Molten zinc chloride

c Sodium iodide solution

d Molten sodium iodide

e Copper(II) chloride solution

f Dilute hydrochloric acid

g Magnesium sulfate solution

h Sodium hydroxide solution

SKILLS PROBLEM SOLVING

2 Copy and complete the following half-equations for reaction at the anode or cathode, and state whether each involves oxidation or reduction. Electrons have been put in for the first four but not for the ones after that.

a $Mg^{2+} + e^- \rightarrow Mg$

b $Al^{3+} + e^- \rightarrow Al$

c $2Br^- \rightarrow Br_2 + e^-$

d $O^{2-} \rightarrow O_2 + e^-$

e $Cl^- \rightarrow Cl_2$

f $Ni^{2+} \rightarrow Ni$

g $OH^- \rightarrow O_2 + H_2O$

h $H_2O \rightarrow O_2 + H^+$

i $H_2O \rightarrow H_2 + OH^-$

SKILLS REASONING

3 Some solid potassium iodide was placed in an evaporating basin. Two carbon electrodes were inserted and connected to a 12 volt DC power source and a light bulb. The potassium iodide was heated. As soon as the potassium iodide was molten, the bulb came on. Purple fumes were seen coming from the positive electrode, and lilac flashes were seen around the negative one.

a Explain why the bulb didn't come on until the potassium iodide melted.

SKILLS CRITICAL THINKING 6

b State the name of the positive electrode.

8

c Name the purple fumes seen at the positive electrode, and write the ionic half-equation for their formation.

d The lilac flashes seen around the negative electrode are caused by the potassium which is formed. The potassium burns with a lilac flame. Write the ionic half-equation for the formation of the potassium.

6

e State the products formed at the electrodes if molten sodium bromide is electrolysed instead of molten potassium iodide.

8

f Write the ionic half-equations for the reactions occurring during the electrolysis of molten sodium bromide.